劉晉安 Chin-An, Liu

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SUMMARY

Master of Computer Science graduate, specialising in Data Science and AI. Focused on **MLOps and full-stack AI platform deployment**. Experienced in optimizing model inference (75% latency reduction) and managing end-to-end model lifecycles. Dedicated to enhance **manufacturing productivity and system scalability** through robust AI solutions and collaboration.

EDUCATION

University of Sydney — Sydney, Australia

Master of Computer Science — Jul. 2024 ~ Nov. 2025

WAM: 76, **GPA:** 3.4

National Pingtung University — Pingtung, Taiwan

Bachelor of Computer Science and Artificial Intelligence — Sep. 2020 ~ Jun. 2024

WAM: 87.1, **GPA:** 3.67

SKILLS

ML & DL: CNN, YOLO, Transformer, PyTorch, TensorFlow, Scikit-learn, Computer Vision (OpenCV), Prompt Engineering

MLOps & Full stack Development: Docker, CI/CD Pipelines, Git, React, Node.js, SQL

Programming Languages: Python, C/C++, C#, Java, R

Languages: English (IELTS Academic 6.5), Mandarin (Native)

WORK EXPERIENCE

AI Engineer (RDSS) — Hsinchu, Taiwan

Vanguard International Semiconductor Corporation (VIS) — Sep. 2026 ~ Present

- Enhanced wafer defect map classification and clustering algorithms, boosting accuracy, efficiency, and overall yield.
- Analysed complex datasets to identify actionable business insights and continuous process improvement opportunities.

Multimedia Research Assistant — Pingtung, Taiwan

National Pingtung University — Oct. 2023 ~ Jul. 2024

- Optimised Reversible Data Hiding (RDH) algorithms to enhance data embedding capacity (EC) by 13%, ensuring image data integrity.
- Synthesized State-of-the-Art (SOTA) research from multiple journals to identify and evaluate optimal methodologies.
- Established a standardized literature review process that improved the precision of experiment reproduction and validation.

AWARDS & CERTIFICATES

Google AI Professional Certificate, App Building, Data Analysis — 2026

書卷獎 — 2022, 2023

PROJECTS

Master's Capstone Project – Automated Detection of Pollinator–Flower Interactions from Citizen Science Images — 2025

- Led a 7-member team through proposal and system development, aligning weekly progress with client requirements.
- Developed and productionized an Automated Floral Feature Recognition System and fine-tuned YOLO and a multi-head Vision Transformer for ROI detection and feature extraction. Achieved over 90% accuracy, providing real-time automated analysis.
- Optimized AI inference pipelines within a Docker-based Full-stack environment, successfully reducing end-to-end latency from 3s to 750ms (75% reduction) to support high frequency real-time analysis and improve system throughput and scalability.

Bachelor's Capstone Project – 結合深度學習與影像加密技術在資訊通訊上安全機制之設計與實作 — 2023

- Developed a Biometric Authentication System integrating Face Recognition, Visual Speech Recognition and RDH algorithms. Improved verification accuracy from 79% to 92% to meet real-time financial verification requirements.